Ahmed Hassan

CS Student | Cybersecurity & SOC Operations

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Profile

Computer Science student at UET Lahore with hands-on experience in network traffic analysis, threat detection, and security monitoring. Experienced in building detection and defense systems, analyzing anomalous network behavior using Python and Wireshark, and identifying OWASP Top 10 vulnerabilities through structured lab environments. Familiar with incident response workflows and alert triage through TryHackMe and CTF platforms. Seeking to contribute to real-world threat monitoring, log analysis, and incident detection in a SOC environment.

Cybersecurity Training

TRYHACKME | Pre-Security & Jr. Penetration Tester Paths

Self-Directed | 2024 -- Present

- Completed 20+ networking and security labs covering threat detection, vulnerability analysis, and incident response workflows.
- Practiced enumeration, traffic analysis, and log investigation using Nmap, Wireshark, Gobuster, and Burp Suite to identify active threats and attack patterns (SQLi, XSS, IDOR).

PICOCTF & OVERTHEWIRE | Web Exploitation, Linux, Forensics

Self-Directed | 2024 -- Present

- Solved CTF challenges in web exploitation, log forensics, privilege escalation, and shell scripting — building an attacker mindset that informs effective threat detection.
- Conducted web enumeration to map attack surfaces and identify hidden endpoints, directly applicable to alert investigation and incident triage.

Technical Projects

NETWORK THREAT DETECTION & DEFENSE SYSTEM | Python, Scapy, Wireshark

Information Security Lab | Phase 1 Completed

- Built a real-time network monitoring tool that captures live traffic, establishes baselines of normal IP-MAC behavior, and raises automated alerts on anomalous ARP activity — simulating core SOC detection logic.
- Demonstrated HTTP/HTTPS interception and HSTS downgrade attacks in a controlled lab, documenting findings with structured incident reports.
- Designed the defender module to continuously monitor network state and escalate spoofing events — directly mirroring alert triage and escalation workflows in a SOC.

ENTERPRISE NETWORK ARCHITECTURE | Cisco Packet Tracer

Academic Project | Spring 2025

- Engineered a segmented enterprise network with RIP, EIGRP, and OSPF, implementing VLANs, ACLs, dynamic NAT, and DHCP security controls — developing deep knowledge of network topology that underpins effective log analysis and incident investigation.

SECURE SOFTWARE DEVELOPMENT PROJECTS | Full-Stack, Desktop, GUI Apps

Academic Projects | 2024 -- 2025

- Applied secure coding practices (parameterized queries, environment variables) to prevent SQLi and API credential exploits

Education

UET LAHORE

B.Sc. Computer Science

Sep 2024 -- Present | 4th Sem | CGPA: 3.57

Skills

SECURITY & MONITORING TOOLS

Wireshark • Nmap • Burp Suite
OWASP ZAP • Nessus • Gobuster
Kali Linux • Metasploit • VirtualBox

NETWORKING

TCP/IP • DNS • HTTP/S • VLANs
NAT • DHCP • Firewalls • ACLs
Cisco Packet Tracer

PROGRAMMING

Python • Bash • C/C++ • C#
SQL • HTML5 • CSS • JS • LaTeX

SOC & SECURITY CONCEPTS

Threat Monitoring & Detection
Log & Traffic Analysis
Alert Triage & Escalation
Incident Response Fundamentals
OWASP Top 10 • SQLi • XSS
Vulnerability Assessment
Reconnaissance • OSINT
Network Segmentation

Certifications

Completed: TryHackMe Pre-Security
In Progress: TryHackMe Jr. Pentester
Preparing for: CompTIA Security+, eJPT

Extracurriculars

Organizing Member | ITEC
Designed problem sets & logistics (100+ pax).

Participant | ITEC & ACM UET
Coding Competitions